placement.py

`src/moe_pruning/model/placement.py · 133 lines · module moe_pruning.model.placement`

What it does

Decides where model weights live: the whole model on one device, or split over several GPUs with accelerate (`DEVICE_MAP`, `MAX_MEMORY`), e.g. OLMoE over Kaggle's 2x T4.

How it works

- `parse_device_map()` / `parse_max_memory()` validate the settings (named maps such as `auto`, or a JSON object; per-device caps such as `0:13GiB,1:13GiB`).
- `from_pretrained_kwargs()` gives the arguments for loading straight onto the devices.
- `place_model()` moves an in-memory model to one device or dispatches it over several (used for pruned checkpoints).
- `module_device()` and `input_device()` let the rest of the code follow each layer's own device instead of assuming a single one.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class ModelPlacement`	Primary device plus optional multi-GPU device map and memory caps.
`.sharded() [property]`	True when the model is split over several devices.
`.describe()`	Human-readable description for logs.
`parse_device_map(value)`	Validate DEVICE_MAP (empty, a named strategy or a JSON object).
`parse_max_memory(value)`	`"0:13GiB,1:13GiB,cpu:20GiB"` -> `{0: "13GiB", 1: "13GiB", "cpu": "20GiB"}`.
`placement_from_config(cfg)`	Build the placement from DEVICE, DEVICE_MAP and MAX_MEMORY.
`from_pretrained_kwargs(placement)`	Keyword arguments for `from_pretrained` (weights go straight to their devices).
`place_model(model, placement)`	Move an in-memory model to its device(s).
`module_device(module)`	Device of a module's first parameter (or buffer).
`input_device(model)`	Device that model inputs must be placed on (the input embedding's).
`model_devices(model)`	All devices holding parameters of the model.

Settings it reads

`DEVICE`, `DEVICE_MAP`, `MAX_MEMORY`

Connections

Uses: `moe_pruning.utils.device`

Used by: `scripts/replace_experts.py`, `src/moe_pruning/config.py`, `src/moe_pruning/evaluation/evaluator.py`, `src/moe_pruning/model/loader.py`, `src/moe_pruning/pruning/replacement.py`, `src/moe_pruning/serialization/checkpoint.py`, `src/moe_pruning/training/bias_trainer.py`